

Defect Identification of Power Line Insulators Based on a MobileViT-Yolo Deep Learning Algorithm

Weidong Zan*, Non-member
 Chaoyi Dong*,**,*a, Non-member
 Zhiming Zhang*, Non-member
 Xiaoyan Chen*,**,*a, Non-member
 Jianfei Zhao****, Non-member
 Fu Hao****, Non-member

Power line insulator defect identification usually suffers from complex backgrounds, small defect target sizes, and inconspicuous defect features. Traditional identification methods based on image processing, image analysis, and pattern classification have many limitations in solving the aforementioned problems. In recent decades, deep learning classification methods have gradually replaced traditional identification methods in the task of power line insulator defect identification. To accurately identify the locations of insulator defects, this paper proposes an insulator defect detection algorithm using an improved lightweight YOLOv4-tiny network (ILYTN). First, the CBL (Conv-BN-LeakyReLU) modules of the backbone network are replaced with MobileViT blocks to enhance the feature extraction capability of the backbone network. Second, coordinate attention (CA) is introduced in the feature fusion part to improve the network's ability to focus on the location of defects. Finally, an EIOU (efficient intersection over union) loss function, instead of the original CIOU loss function, is used so that the convergence speed of the network can be improved. To verify the effectiveness of the proposed algorithm in this paper, the mViT-yolo algorithm is compared with the mainstream Faster-RCNN algorithm, SSD algorithm, YOLOv3 algorithm, and YOLOv4-tiny algorithm. The experimental results show that the algorithm proposed in this paper outperforms all the above algorithms in terms of detection accuracy. Compared with the traditional YOLOv4-tiny algorithm, the proposed algorithm increases the mean average precision (mAP) by 1.64%, the average precision (AP) of missing insulator defects by 0.32%, and the average precision (AP) of broken insulator defects by 4.96%. © 2023 The Authors. *IEEJ Transactions on Electrical and Electronic Engineering* published by Institute of Electrical Engineer of Japan and Wiley Periodicals LLC

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1. Introduction

The development of smart grids has always been a focus of the Chinese government and power industry. In the field of transmission line inspection, it has gradually become crucial to simplify the inspection process and improve inspection efficiency. Insulators, as important devices to support cable and electrical isolation, are subject to long-term environmental erosion. Therefore, most insulators will inevitably develop defects such as self-explosion

and breakage [1]. These defects can seriously affect the stable operation of transmission lines and even bring about very large economic losses.

To achieve a safer and more stable operation of power production, transmission line inspections increasingly require a lower fault misdiagnosis rate, higher intelligence, and less time and manpower consumption [2]. In the past, transmission line inspection methods mainly used manual proximity at a short range to find the locations of defects. This method usually requires that inspectors have strong professional knowledge, and hazardous conditions can easily occur. Therefore, its inspection efficiency was low [3]. In recent years, with the rapid development of drone applications, an increasing number of power companies are using drone inspection methods to find defects (see Fig. 1); hence, transmission line inspections are gradually moving toward a digital and intelligent era. Initially, the images captured by UAVs were processed by some traditional detection algorithms, e.g., threshold segmentation [4] and spatial morphological feature methods [5]. However, these traditional detection algorithms require high target background divisibility, have complex detection processes, have

^a Correspondence to: Chaoyi Dong. E-mail: dongchaoyi@imut.edu.cn

*College of Electric Power, Inner Mongolia University of Technology, Hohhot 010080, China

**Intelligent Energy Technology and Equipment Engineering Research Centre of Colleges and Universities in Inner Mongolia, Hohhot 010080, China

***Engineering Research Center of Large Energy Storage Technology, Ministry of Education, Hohhot 010051, China

****New Energy Branch Company, Inner Mongolia Electric Power Survey & Design Institute Co., Ltd., Hohhot 010020, China



Fig. 1. UAV during a site inspection operation

low recognition accuracy rates, and are prone to problems such as missed and false detections. To tackle the above problems, deep learning technology has gradually provided new ideas for power line and power equipment inspections [6].

In recent years, researchers have performed a great deal of work related to insulator and insulator defect detection. Reference [7] identified the locations of insulator defects based on the differences in spatial morphological features and color features between insulators and insulator defects. However, this method encountered difficulty in segmenting insulators from complex backgrounds, thus resulting in a significant reduction in detection accuracy. Li *et al.* directly obtained insulator locations by a contour projection method and identified insulators by an SVM (support vector machine) classifier [8]. This method used fewer samples and had a lower generalization ability; therefore, it obtained a low detection accuracy. In reference [9], a cascaded depth segmentation network was used to extract insulators, followed by a DefGAN model to detect defects. This method solved the problem of sample sparsity, but the whole detection process is too complicated and usually consumes large amounts of time and memory. Other researchers used a Faster-RCNN model [10] and the improved Faster-RCNN model [11] to detect self-explosion defects of insulators. Although the detection accuracies of the Faster-RCNN algorithms were improved, there were still some cases of missing detections. Gao *et al.* used an improved YOLOv4 network model to identify self-explosion defects of insulators, which improved the defect detection accuracy to a level of 88.07% [12]. However, the number of network model parameters was large, and the training and reasoning time of the model was long; thus, the detection speed was still unsatisfactory. Chen *et al.* used an INSU-YOLO model combined with an attention mechanism to detect insulator defects, which improved the low recognition rate of small target defects [13]. However, the data set was only limited to glass insulators, and no further study was conducted on other types of insulators.

Previous studies have shown that insulator defects can be detected at different effectiveness degrees, but the accuracy and speed of detection still need to be improved. In addition, most studies have only focused on missing insulator defects. The types of insulator defects investigated are limited, and other types of insulator defects, e.g., broken insulator defects, are rarely considered. To address these problems, this paper addresses both missing insulator defects and broken insulator defects and proposes an improved lightweight YOLOv4-tiny network (ILYTN) to detect the locations of these insulator defects. First, a MobileViT block is introduced in the backbone feature extraction part of the YOLOv4-tiny network. The MobileViT block effectively combines the advantages of inductive bias of convolution with the global perceptual field capability of a transformer, thus enhancing the

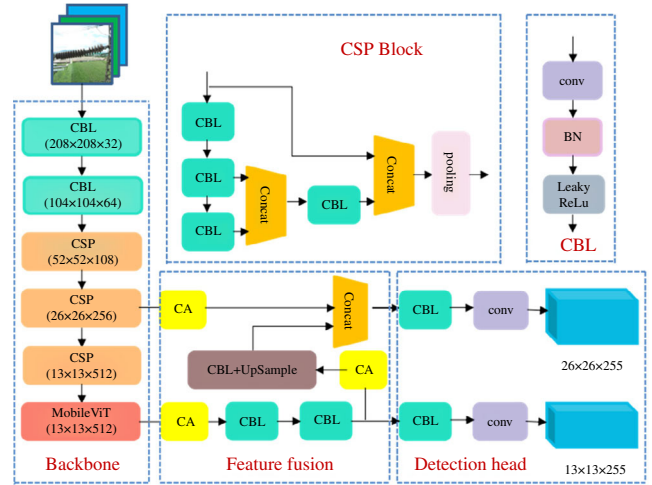


Fig. 2. Structure of the improved lightweight YOLOv4-tiny network

feature extraction capability of the original network. Second, one type of attention module, coordinate attention (CA), is added to the feature fusion part of the YOLOv4-tiny network to increase the accuracy of locating the target of interest. Finally, an improved loss function is employed to replace the previous loss function, CIOU, of the YOLOv4-tiny network, which further accelerates the convergence speed and improves the regression accuracy. The experimental results demonstrate that the proposed ILYTN model is superior to the Faster-RCNN, single-shot multi-box detector (SSD), and YOLOv3 models in terms of insulator defect detection accuracy. Compared with the traditional YOLOv4-tiny deep learning algorithm, the feature extraction performance of ILYTN is improved by 1.64% in mean average precision (mAP), 0.32% in average precision (AP) for missing insulator defects, and 4.96% in AP for broken insulator defects.

2. Methods

2.1. ILYTN structure Because the YOLOv4-tiny network model has a smaller parametric number and is easier to deploy on embedded devices, this network model can be viewed as a lightened version of the original YOLOv4 network model [14]. Based on the YOLOv4-tiny network model, we further proposed an improved network structure, ILYTN, whose structure is shown in Fig. 2.

ILYTN consists of three parts: a backbone feature extraction subnetwork, a feature fusion subnetwork, and the detection head subnetwork. The backbone feature extraction subnetwork consists of two CBL modules, three CSP modules, and a MobileViT module, which downsample input images and extract target features. Then, the feature fusion subnetwork merges the outputs of two feature layers from the backbone feature extraction network. This subnetwork processes both the location and detail information of the lower layer features and the semantic information of the higher layer features together, thus greatly guaranteeing excellent model performance. The prediction head subnetwork uses a nonmaximum suppression algorithm to select multiple bounding boxes of the same category. In the following, the target categories can be predicted, while the bounding boxes are generated.

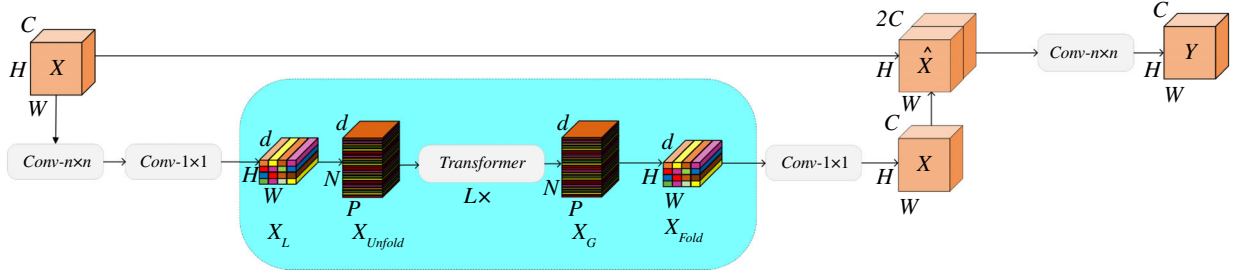


Fig. 3. MobileViT module

2.2. MobileViT block With the development of deep learning technology in recent years, transformer networks [15,16] have gradually been transferred from the well-performing NLP (natural language processing) domain to the CV (computer vision) domain. Compared with traditional convolutional networks, such as ViT [17] and DETR [18], transformer networks have better performance in vision tasks. However, transformer network structures usually have a larger number of parameters, therefore consuming a great deal of computational effort. This MobileViT module effectively combines the inductive bias property of convolution neural networks with the global receptive field capability of transformer networks in the following way [19]:

For a given input $X \in \mathbb{R}^{H \times W \times C}$ (where H and W represent the height and width of the feature map and C represents the number of channels), the MobileViT module first learns the local spatial information using an $n \times n$ convolution, and then a tensor is projected into a d -dimensional space using a 1×1 convolution to obtain $X_L \in \mathbb{R}^{H \times W \times d}$, where $d > C$. Next, X_L is expanded into N blocks of pixels to obtain $X_{\text{Unfold}} \in \mathbb{R}^{P \times N \times d}$ and $P = h \times w$. Notations h and w represent the width and height of the pixel blocks, respectively, and $N = (H \times W)/P$ is the number of pixel blocks. Then, a transformer module is used to encode the global information of the pixel at the p th position in the block of P pixels to obtain $X_G \in \mathbb{R}^{P \times N \times d}$, as shown in (1).

$$X_G(p) = \text{Transformer}(X_{\text{Unfold}}(p)), 1 \leq p \leq P \quad (1)$$

To avoid losing the sequence between the pixel blocks, $X_G \in \mathbb{R}^{P \times N \times d}$ is collapsed to obtain $X_{\text{Fold}} \in \mathbb{R}^{H \times W \times d}$. Then, a 1×1 convolution is used to project X_{Fold} into the C -dimensional space and cascade it with the input X to obtain augmented $\hat{X} \in \mathbb{R}^{H \times W \times 2C}$. Finally, an $n \times n$ convolution fuses the local and global features to obtain the output $Y \in \mathbb{R}^{H \times W \times C}$. The previously described MobileViT module is shown in Fig. 3.

Figure 4 illustrates the reason why the MobileViT module has the ability to learn global information and thus can improve detection accuracy. The black grids in Fig. 4 represent a block of pixels, and the gray grids represent a pixel. The red pixel uses the Transformer module to process its surrounding eight blue pixels, each of which already encodes neighboring pixel information through convolution operations. In this way, the red pixel can encode all the pixels in the image.

2.3. CA module The CA structure is shown in Fig. 5 [20], and it is divided into two steps: 1. The first stage is coordinate information embedding. For a given input $X = [x_1, x_2, \dots, x_C] \in \mathbb{R}^{H \times W \times C}$, two spatial pooling kernels ($H, 1$) and ($1, W$) are used to perform the pooling operations in the horizontal and vertical

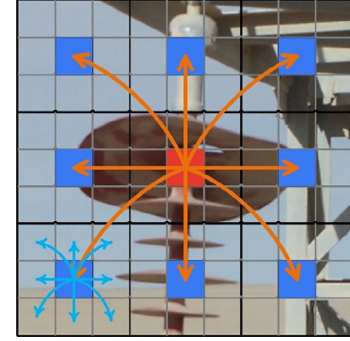


Fig. 4. Image processing of a MobileViT module

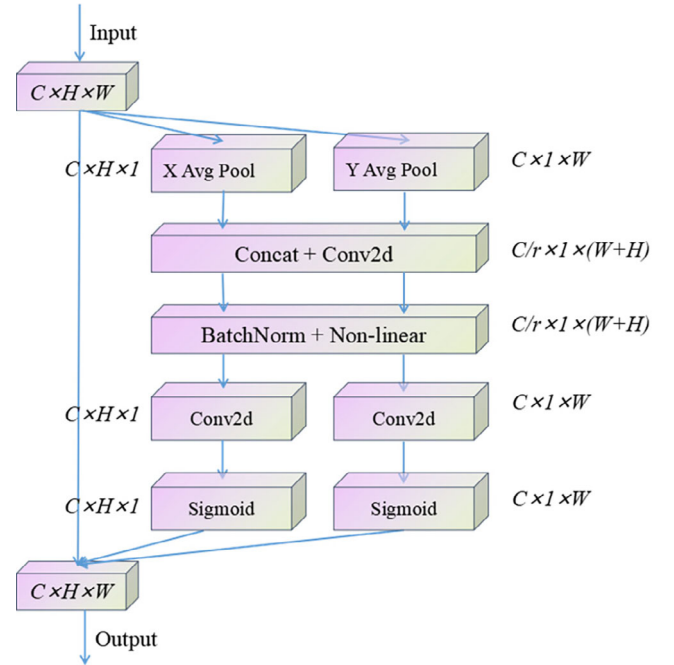


Fig. 5. CA structure

directions, respectively. The two operations encode each channel to maintain the coordinate information of spatial locations. The specific process is shown in (2) and (3).

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leq i \leq W} x_c(h, i) \quad (2)$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leq j \leq H} x_c(j, w) \quad (3)$$

where $x_c(h, i)$ is the pixel location information of feature x_c at height h and width i , $x_c(j, w)$ is the pixel location information of feature x_c at height j and width w , and $z_c^h(h)$ and $z_c^w(w)$ denote the coordinate perception attentional feature maps along the horizontal and vertical directions on channel c , respectively. 2. The second stage is coordinate attention generation. The feature maps generated by (2) and (3) are concatenated and transformed by a 1×1 convolution function F_1 :

$$f = \delta(F_1([z^h, z^w])), \quad (4)$$

where δ is a nonlinear activation function, $[\cdot]$ is a dimension catenation operation, and $f \in \mathbb{R}^{C/r \times (H+W)}$ denotes the feature map after encoding the spatial information in the horizontal and vertical directions, where r is the downsampling ratio. Then, the feature map f is divided into two separate tensors f^h and f^w . The two tensors are convoluted using two 1×1 convolution operations F_h and F_w and then transformed using two *Sigmoid* activation functions. See (5) for details:

$$\begin{aligned} g^h &= \text{Sigmoid}(F_h(f^h)) \\ g^w &= \text{Sigmoid}(F_w(f^w)) \end{aligned} \quad (5)$$

where g^h and g^w denote the attention weights generated in the horizontal and vertical directions, respectively. Finally, these generated attention weights are applied to the input features to obtain an output y_c :

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j). \quad (6)$$

A traditional channel attention mechanism uses global average pooling to encode empty information, and the location information is difficult to retain after global average pooling compression. The CA mechanism, however, adds location information to the channel attention mechanism, enabling the network model to capture not only channel information but also location information. Since it is equivalent to the feature selection mechanism introduced in traditional machine learning, the introduction of the attention mechanism simplifies the model structure and improves the generalization ability and test accuracy of ILYTN.

2.4. Improved loss function A loss function is used in the YOLOv4-tiny detection algorithm to represent the degree of disparity between the prediction and the actual data. The selection of a suitable loss function not only significantly influences the convergence speed during model training but also the accuracy of target detection. The *IOU* function is commonly used in image detection to measure this degree of disparity [21], and it is defined as:

$$L_{IOU} = 1 - IOU = 1 - \left| \frac{A \cap B}{A \cup B} \right| \quad (7)$$

In (7), notation A represents the area of the real frame, and notation B represents the area of the predicted frame. When $A \cap B = 0$, that is, when the real frame and the predicted frame do not intersect, the value of *IOU* is always 0. At this time, there is no gradient return, and the learning cannot be continued. However, when the *IOU* values are the same, there are situations in which the positional relationship between the real frame and the predicted frame

cannot be determined. Therefore, the YOLOv4-tiny network uses another loss function, the CIOU loss function [22], which takes into account the overlapping area, centroid distance, and width-height ratio of the prediction frame and the real frame. The CIOU is defined in (8)–(10).

$$L_{CIOU} = 1 - IOU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (8)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (9)$$

$$\alpha = \frac{v}{(1 - IOU) + v} \quad (10)$$

where $\rho^2(b, b^{gt})$ represents the Euclidean distance between the center point of the predicted frame and that of the true frame; c represents the diagonal length of the smallest external bounding rectangular frame encompassing the predicted frame and the true frame; v is a parameter measuring the consistency of the width-height ratio; w and h are the width and the height of the predicted frame, respectively; w^{gt} and h^{gt} are the width and the height of the true frame, respectively; and α is a trade-off parameter between the two penalty items. Nevertheless, this width-height ratio calculation in CIOU can only represent a fuzzy difference between the real frame and the predicted frame and is not the actual error value between the two frames. Based on this consideration, this paper uses an EIOU loss function [23] to replace the CIOU loss function. The EIOU loss function formula is described as follows:

$$L_{EIOU} = 1 - IOU + \frac{\rho^2(b, b^{gt})}{c^2} + \frac{\rho^2(w, w^{gt})}{c_w^2} + \frac{\rho^2(h, h^{gt})}{c_h^2} \quad (11)$$

where $\rho^2(w, w^{gt})$ is the Euclidean distance between the width of the predicted frame and that of the real frame; $\rho^2(h, h^{gt})$ is the Euclidean distance between the height of the predicted frame and that of the real frame; and c_w and c_h are the width and the height of the smallest external bounding rectangular frame encompassing the predicted frame and the real frame, respectively. Since the EIOU loss function calculates the differences in width and height separately instead of the width-height ratio, it further accelerates the convergence speed and improves the regression accuracy.

3. Experiment

3.1. Data sets the insulator image data sets used in this study include two parts

(i) 248 images of defective insulators from the China power line insulator data set (CPLID), which are publicly available on the internet, and (ii) 389 images of insulators taken by the Northern Airlines Company of the Inner Mongolia Autonomous Region (ACIMAR) using an unmanned aerial vehicle. Due to the small numbers of images in the data sets, the network model is prone to overfitting during the training process. Therefore, this paper uses an augmentation algorithm to expand the data sets; the augmentation operations include random image flipping, color transformation, and adding Gaussian noise. During augmentation, we artificially added noise or alterations to the images to mimic more changes in the environment. For example, a blurred image is formed by adding noise, and a snowy scene is generated by a color change in the background. However, since all original images in our data set were captured by actual fault inspection operations of UAVs from Inner Mongolia Northern



Fig. 6. Marking a defect insulator image as one sample of the training data set using 'Labelimg'

Table I. Running environment of the ILYTN algorithm

Configuration	Parameters
Operating system	Windows 10 64-bit
CPU	Intel(R) Xeon(R) Gold 5218R CPU @ 2.10 GHz
GPU	NVIDIA GeForce RTX3090
Accelerated environment	CUDA 11.0, CUDNN 8.0
Development tools	Pycharm2021

Airlines, the data set naturally incorporated the environmental disturbances, for example, weather changes, vibration blur, and motion blur. The data sets are randomly divided into a training set, a validation set, and a test set at a ratio of 8:1:1. These images are annotated by the image annotation tool 'Labelimg'. For example, in Fig. 6, an intact insulator is marked as 'insulator', a missing insulator is marked as 'defect', and a damaged insulator is marked as 'broken'.

3.2. Experimental setting The experimental environment for the proposed ILYTN algorithm is shown in Table I.

3.3. Evaluation criteria To accurately analyze the performance of the model, many evaluation criteria for predictions, for example, the category average precision (AP), mean category average precision (mAP), precision (P), recall rate (R), and frames per second (FPS) metrics, are used to evaluate the effectiveness of predicting insulator defects. Equations (12) and (13) are used to obtain the values of precision and recall.

$$P = \frac{TP}{TP + FP} \quad (12)$$

$$R = \frac{TP}{TP + FN} \quad (13)$$

where TP (true positive) indicates the actual number of positive samples predicted to be positive, FP (false-positive) indicates the

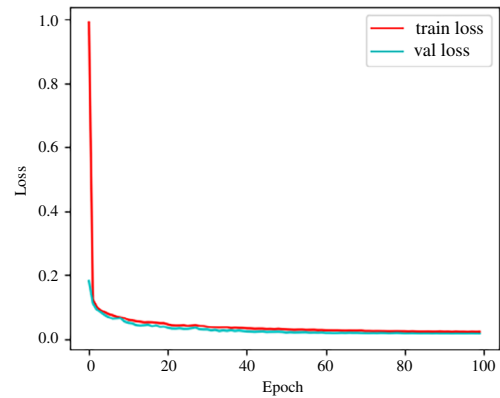


Fig. 7. Training loss variation

actual number of negative samples predicted to be positive, and FN (false-negative) indicates the actual number of positive samples predicted to be negative.

3.4. Model training This experiment trains the deep learning networks using the expanded data sets in a PyTorch environment. The specific experimental parameters are set as follows: the batch size is set to 64, the Adam optimization strategy is used, the learning rate is initialized to 0.001, and the network model is trained for a total of 100 epochs. The changes in its training loss function are shown in Fig. 7, where 'train loss' is the training set loss and 'val loss' is the validation set loss. The figure shows that as the number of training epochs of the model increases, the training set loss curve gradually decreases until it tends to plateau. The validation set loss curve also has no rising trend, and the model does not experience an overfitting phenomenon.

To verify the effectiveness of the proposed ILYTN method, the performance of ILYTN is compared with that of previous networks using both the images from the public data set, CPLID, and the field ACIMAR data set.

Table II. A comparison of the detection results of the MobileViT block and the conventional transformer block

Classification performance	Insulator AP/%	Defect AP/%	Broken AP/%	mAP/%	Speed/FPS
Conventional transformer block	98.65	99.28	84.98	94.30	63.23
MobileViT block	97.01	98.72	87.28	94.34	80.61

Table III. Effect of the size of block P on the detection results

P size	Insulator AP/%	Defect AP/%	Broken AP/%	mAP/%	Speed/FPS
P = 1	97.01	98.72	87.28	94.34	80.61
P = 4	96.97	98.13	83.07	92.72	81.32
P = 16	96.51	95.94	81.57	91.34	83.23
P = 64	94.84	95.03	78.57	89.48	84.78

Table IV. MobileViT block and dilated convolution block comparison results

Classification performance	Insulator AP/%	Defect AP/%	Broken AP/%	mAP/%	Speed/FPS
Dilated convolution	97.09	99.05	81.80	92.65	90.64
MobileViT block	97.01	98.72	87.28	94.34	80.61

3.5. Experimental results

3.5.1. Comparison of experimental results To verify the detection effectiveness of the MobileViT block, we compare it with the conventional transformer block. The different performances of the two types of blocks are shown in Table II:

As seen from Table II, the conventional transformer block slightly outperforms the MobileViT block in terms of insulator detection accuracy and missing defect detection accuracy compared to the MobileViT block. On the other hand, the MobileViT block has a 2.3% higher broken defect detection accuracy than the conventional transformer block. The class average accuracy of the two blocks is basically the same. However, the detection speed of the conventional transformer block is significantly lower than that of the MobileViT block due to the large number of network model parameters in the conventional transformer block.

In the MobileViT block, the effect of the size of the pixel block P on the experimental performance is analyzed, as shown in Table III.

Table III shows that as the patch size increases, the network model loses more information in the training process, and the accuracy of both insulator detection and insulator defect detection gradually decreases. However, the detection speed of the network model gradually increases with increasing patch size.

The results of the MobileViT block compared to the expanded convolution block are shown in Table IV.

As seen in Table IV, the dilated convolution block performs better than the MobileViT block in terms of the detection accuracy of insulators and missing defects. However, its broken defect detection accuracy is 5.48% lower than that of MobileViT, and its average accuracy is 1.69% lower than that of the MobileViT block. The dilated convolution block outperforms the MobileViT

Table V. Ablation experiments with different improved strategies

Experiment index	YOLOv4-tiny	MobileViT block	CA	EIOU	mAP/%	Speed/FPS
1	✓				92.70	103.22
2	✓	✓			93.39	81.99
3	✓		✓		93.22	85.97
4	✓			✓	93.09	113.23
5	✓	✓	✓		93.96	81.65
6	✓	✓	✓	✓	94.34	80.61

Table VI. Performance comparison of the different deep learning models

Model	Insulator AP/%	Defect AP/%	Broken AP/%	mAP/%	Speed/FPS
Faster-RCNN	97.45	62.27	62.56	74.09	22.55
SSD	94.23	78.10	72.95	81.76	89.59
YOLOv3	95.50	79.22	74.27	83.00	93.17
YOLOv4-tiny	97.39	98.40	82.32	92.70	103.22
ILYTN	97.01	98.72	87.28	94.34	80.61

block in terms of detection speed due to its fewer network model parameters and less computational effort.

To analyze the impact of different improvement components on the performance of the ILYTN model, six sets of experiments are designed and performed using the same test set. The test results are shown in Table V, where the notation '✓' indicates the strategy used to improve the model. Two types of performances, mAP and FPS, are evaluated in Table V.

Experiment 2 adds the MobileViT block to the backbone network on the basis of Experiment 1 so that the CNN and transformer can be fused and the improved backbone network can increase the global sensing field and feature extraction capability. Thus, the mAP increases by 0.69%. Experiment 3 introduces the CA mechanism based on Experiment 1 to enable the network to capture not only channel information but also location information, thus improving the small target detection accuracy of insulator defects with an mAP improvement of 0.52%. Experiment 4 improves the regression accuracy by replacing the CIOU loss function with EIOU, resulting in a 0.39% improvement in mAP. Experiment 5 uses a fusion of the MobileViT module and CA strategies to enable the network model to locate targets of interest more accurately and learn more features while suppressing features that are not relevant to the targets. Experiment 5 results in a 1.26% improvement in mAP. Experiment 6 fuses all three improved strategies to improve the mAP by 94.34%, which is 1.64% higher than that of Experiment 5. For the detection speed aspect, it can be seen that the EIOU loss function plays a large role in improving the detection speed, making the network model maintain a high detection speed while improving the detection accuracy.

3.5.2. Model comparison To verify the effectiveness of the ILYTN model, we compared the ILYTN model with the other traditional detection models, for example, Faster-RCNN, SSD, YOLOv3, and YOLOv4-tiny. The experimental results are shown in Table VI.

Table VI shows that the Faster-RCNN model performs poorly compared with the other detection models in terms of both the

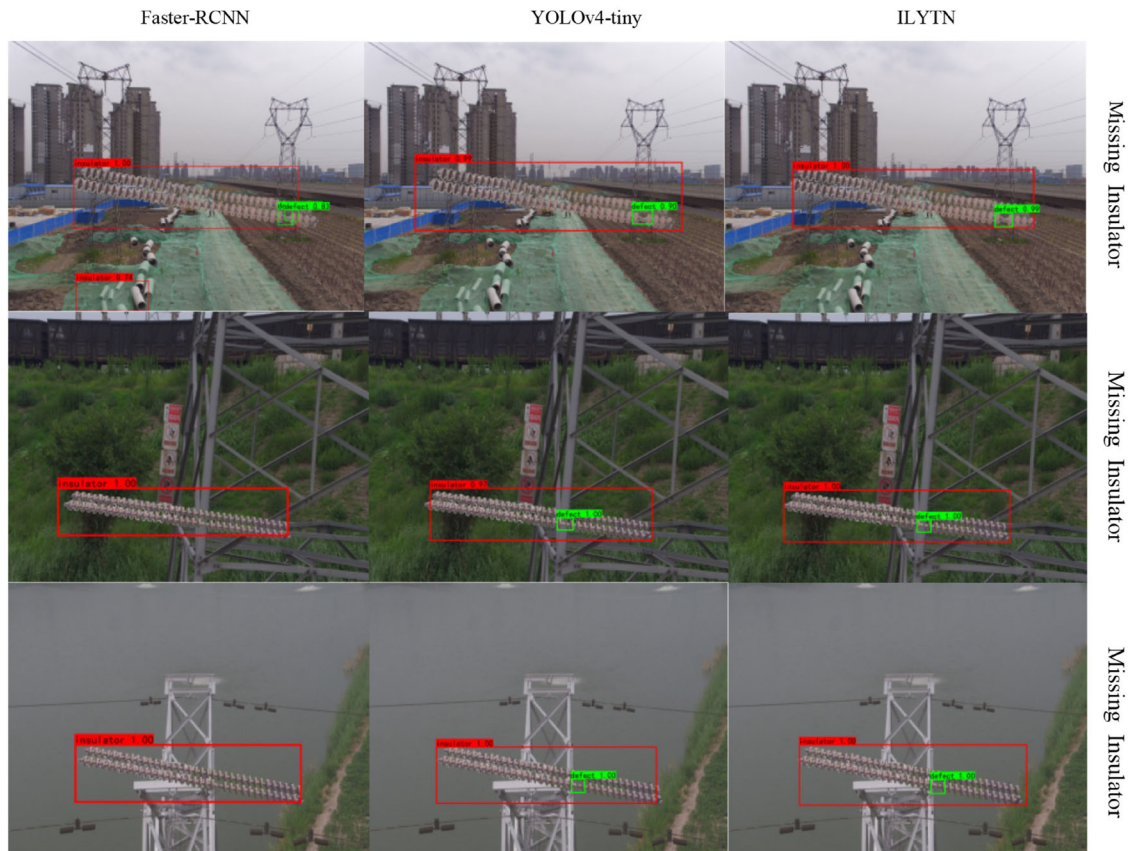


Fig. 8. Illustration of an insulator defect from the public CPLID detected by the three different network models

detection speed and detection accuracy. The SSD network model and the YOLOv3 network model perform well in detecting large targets such as insulators but have relatively low AP values for detecting small targets such as insulator defects and insulator breakages. The ILYTN model has the best precision performance in defect AP, broken AP, and mAP. Compared to the original YOLOv4-tiny network model, it improves the detection accuracy by 0.32% for defective insulators and 4.96% for broken insulators, and it improves mAP by 1.64%. Because of its simplified structure, the detection speed of YOLOv4-tiny is faster than that of the other four networks, achieving 103.22 FPS.

To further demonstrate the superiority of the ILYTN model, the results are shown in Fig. 8 and Fig. 9. As shown in Fig. 8, the three network structures generally perform well in detecting large targets, such as insulators, on the public CPLID. However, in some cases, the Faster-RCNN model is prone to false detections for excessively complex backgrounds. Both the proposed ILYTN model and YOLOv4-tiny network model are good at detecting the locations of defects, and the improved network model has a higher confidence level in detecting defects than the YOLOv4-tiny network model.

The defect detection results of the insulator images from the field ACIMAR data set are shown in Fig. 9. The first and second rows are the images of insulators with missing defects, and the third and fourth rows are images of insulators with broken defects. It can be seen from the first two rows that when the background color is similar to the insulator color, large targets such as insulators can be detected well, while small target defects tend to be incorrectly

detected. When the background is a green field, all three network models are able to detect the defect locations. In the third and fourth rows of images, the same problem as that in the first row exists. The lack of distinctive features for broken defects exacerbates the problem of missing small target defects. Compared to the Faster-RCNN and YOLOv4-tiny models, the proposed ILYTN model in this paper improves the detection accuracy of small target defects with a higher confidence level.

4. Conclusion

We propose an ILYTN insulator defect detection algorithm based on the previous YOLOv4-tiny algorithm to solve the problems caused by the small target sizes of some specific insulator defects, complex backgrounds, and a low recognition rate of the defect category. First, the CBL module at the end of a typical YOLOv4-tiny backbone network is replaced with a MobileViT block, where the former CNN is combined with a transformer to enhance the feature extraction capability of the network. Second, the CA mechanism is added to the feature fusion part to make the network focus more on the target region of interest. Finally, the CIOU loss function is replaced by the EIOU loss function to speed up the convergence of the network and improve the regression accuracy.

As the defect and broken parts usually occupy fewer pixels in the whole frame of insulators, the target sizes of these two types of faults are small. With the increase in the number of deep learning network layers, the feature information of these small

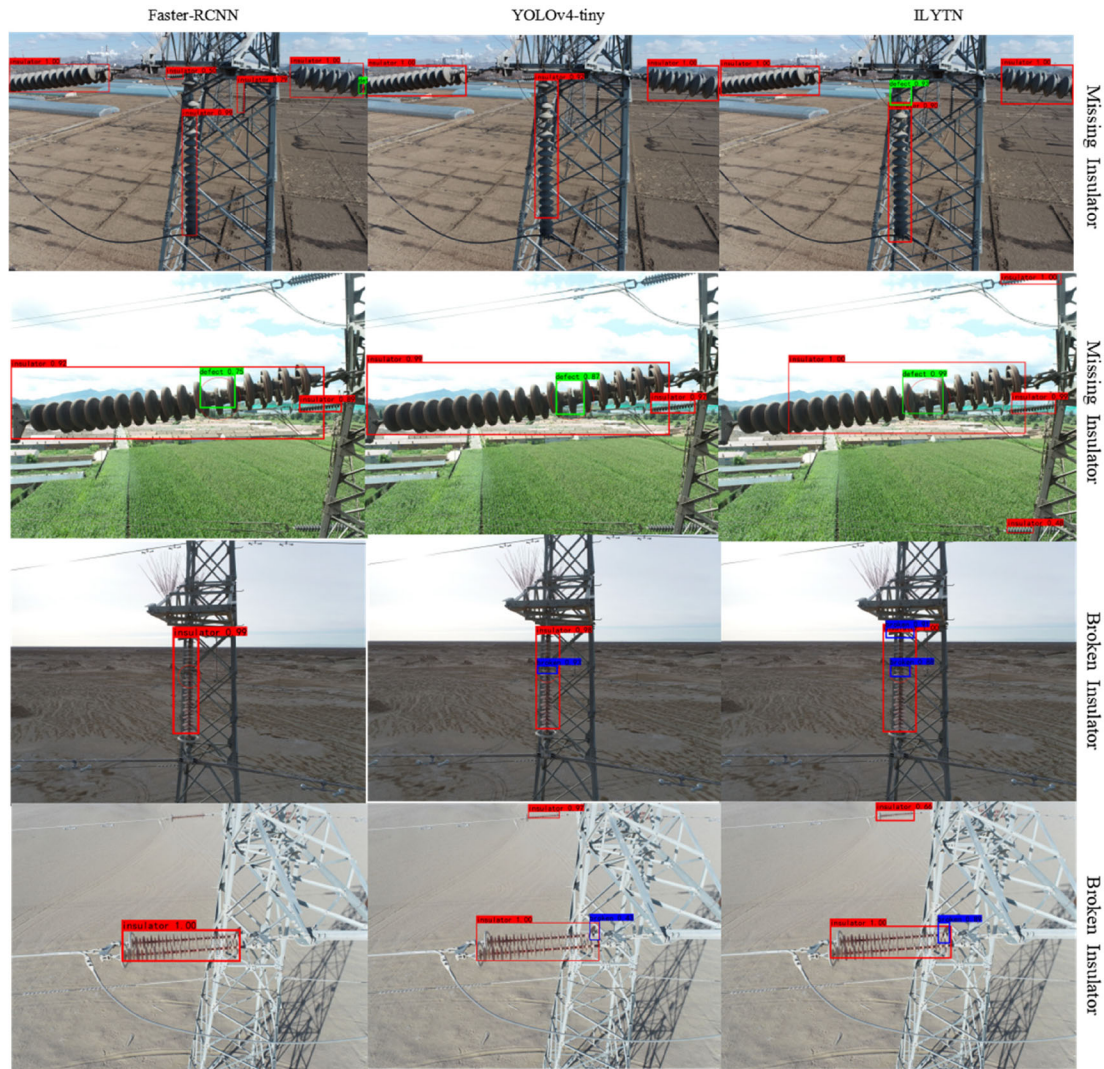


Fig. 9. Illustrations of missing or broken insulator defects from the field ACIMAR data set by the three different network models

targets is gradually lost, making them difficult to detect. In contrast, the YOLOv4-tiny network model has two detection heads, which combine shallow representation information for multiscale learning with deep semantic information of small targets. This effectively improves the small target detection performance of the YOLOv4-tiny network. Furthermore, the proposed method based on the YOLOv4-tiny network enhances the small-target feature extraction capability by introducing the MobileViT block and the attention mechanism. Consequently, YOLOv4-tiny method and the proposed method based on YOLOv4-tiny have significantly improved detection performance for defects and broken insulators.

In the experiments, we compare the ILYTN algorithm with other mainstream algorithms. Compared with the traditional YOLOv4-tiny network model, the mAP of ILYTN increases by 1.64%, the missing insulator AP value of ILYTN improves by 0.32%, and the broken insulator AP value of ILYTN improves by 4.96%. Because of its simplified structure, the detection speed of the traditional YOLOv4-tiny is faster than that of the other four networks, and it is 103.22 FPS. The speed of ILYTN is 80.61 FPS, which is also acceptable for real-time applications. Therefore, the result shows that the ILYTN algorithm achieves the highest detection

precision and maintains a speed that is acceptable for engineering applications. The ILYTN model can meet the engineering demands of the intelligent inspection of transmission lines in practical power production. Future research may focus on how to deploy the proposed network model on a GPU-based master control device to realize the reasoning process of insulator defect detection.

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Weidong Zan (Non-member) received a B.S. degree from Inner Mongolia University of Technology, Hohhot, China, in 2017. He is a master's student at the School of Electric Power, Inner Mongolia University of Technology. His current research interests include computer vision and object detection.



Chaoyi Dong (Non-member) received his Ph.D. from Korea University, Korea, in 2009. He is currently a professor and a master's student supervisor at School of Electric Power, Inner Mongolia University of Technology. His research field includes computer vision, brain-machine interfaces, and robot navigation and control.



Zhiming Zhang (Non-member) received a BS degree from Inner Mongolia University of Technology, Hohhot, China, in 2019. He is a master's student at the School of Electric Power, Inner Mongolia University of Technology. His current research interests include computer vision and object detection.



XiaoYan Chen (Non-member) received her master's degree from Inner Mongolia University of Technology, China, in 2009. She is currently an associate professor and a master's student supervisor at the School of Electric Power, Inner Mongolia University of Technology. Her research field includes control engineering and pattern recognition.



Jianfei Zhao (Non-member) received his Master's degree from Inner Mongolia University of Technology, Hohhot, China in 2016. He is currently a senior engineer at the Inner Mongolia Electric Power Survey & Design Institute Co., Ltd. His research interests include new energy equipment operation and maintenance and electrical control.



Fu Hao (Non-member) received his BS degree from Inner Mongolia Agricultural University in Hohhot, China, in 2010. He is currently an engineer at the Inner Mongolia Electric Power Survey & Design Institute Co., Ltd. His research interests include new energy equipment operation and maintenance and electrical control.

